SUBCHAPTER 3.1 · FUNDAMENTAL KINEMATICS OF INTEGRAL SYSTEMS

The Reynolds Transport Theorem (RTT) & Control Volume Selection Strategy

Converting Lagrangian material derivatives into Eulerian flux balances across fixed, moving, and deformable control boundaries.

1. The Reynolds Transport Theorem Master Formula

In continuum mechanics, physical conservation laws (conservation of mass, Newton's second law, and the first law of thermodynamics) are fundamentally formulated for a **closed material system**: a fixed collection of fluid particles enclosed within a time-dependent, deforming material volume $V_f(t)$ bounded by surface $\Sigma_f(t)$ that moves along with the local fluid velocity $\vec{v}$.

Let B_{sys} denote any extensive property of this closed system (such as mass m, linear momentum $m\vec{v}$, angular momentum $\vec{H}_{\vec{x}_0}$, or total energy E), and let $b = dB/dm$ be the corresponding intensive property per unit mass. The total extensive property stored in the fluid volume is:

$$B_{\text{sys}}(t) = \int_{V_f(t)} \rho b \, dV$$

[Eq. 3.1] Total extensive property B_{sys} as the volume integral of volumetric density $\phi = \rho b$.

To differentiate this integral with respect to time when both the integrand ρb and the spatial domain $V_f(t)$ vary simultaneously, we apply the formal analytical limit definition:

$$\frac{dB_{\text{sys}}}{dt} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \left[\int_{V_f(t+\Delta t)} \rho b(\vec{x}, t+\Delta t) \, dV - \int_{V_f(t)} \rho b(\vec{x}, t) \, dV \right]$$

[Eq. 3.2] Analytical limit definition of the material time derivative.

Over the infinitesimal time interval Δt, the material boundary moves with fluid velocity $\vec{v}$. An elementary surface patch $d\sigma$ sweeps out a cylindrical volume of base $d\sigma$ and height $(\vec{v}\Delta t) \cdot \vec{n}$, yielding $dV = (\vec{v} \cdot \vec{n})\Delta t \, d\sigma$. Splitting the limit into local unsteadiness and boundary convective sweeping yields the classical RTT for a material volume:

$$\frac{d}{dt} \left[\int_{V_f(t)} \rho b \, dV \right] = \int_{V_f(t)} \frac{\partial (\rho b)}{\partial t} \, dV + \int_{\Sigma_f(t)} \rho b (\vec{v} \cdot \vec{n}) \, d\sigma$$

[Eq. 3.6] Reynolds Transport Theorem for a material fluid volume $V_f(t)$.

Universal Extension to an Arbitrary Moving Control Volume $V_c(t)$

In aerospace propulsion and aerodynamics, tracking material fluid packages downstream of a jet engine is impossible. Instead, we define a geometric **control volume** $V_c(t)$ whose control surface $\Sigma_c(t)$ moves with an arbitrary, specified velocity field $\vec{v}_c(\vec{x}, t)$. Matching $V_c(t)$ with $V_f(t)$ instantaneously at time t yields the **Universal RTT Master Formula**:

$$\mathbf{\frac{dB_{sys}}{dt} = \frac{d}{dt}\left[\int_{V_c(t)} \rho b \, dV\right] + \int_{\Sigma_c(t)} \rho b[(\vec{v} - \vec{v}_c)\cdot\vec{n}]\, d\sigma}$$

[Eq. 3.8] Universal Reynolds Transport Theorem with relative velocity $(\vec{v} - \vec{v}_c)$.

> PHYSICAL INTERPRETATION OF THE TWO TERMS
>
> **1. Rate of Storage / Unsteadiness:** $\frac{d}{dt}\int_{V_c} \rho b \, dV$ measures how much of property B accumulates or depletes inside the instantaneous geometric boundaries of $V_c(t)$.
>
> **2. Net Convective Flux:** $\int_{\Sigma_c} \rho b[(\vec{v} - \vec{v}_c)\cdot\vec{n}]\, d\sigma$ accounts for property B carried across the boundary by the *relative velocity* $\vec{w} = \vec{v} - \vec{v}_c$ with which fluid crosses the control surface.

Step-by-Step Derivation of the Reynolds Transport Theorem (Notes.pdf, Eqs. 3.2-3.8)

This section expands every step that the compact derivation above skips. Throughout, $\phi(\vec{x}, t)$ is the volumetric density of the extensive property (in the notation $B = \int \rho b \, dV$ used elsewhere, $\phi = \rho b$). It is assumed twice continuously differentiable in space and time, and the surfaces $\Sigma_f(t)$ and $\Sigma_c(t)$ are smooth and closed, with $\vec{n}$ the outward unit normal. Equation numbers are those of Notes.pdf, Chapter 3.

Step 1: Split the difference quotient (Eqs. 3.2 and 3.3)

The material volume at $t + dt$ is the volume at t plus the thin shell $\delta V = V_f(t + dt) - V_f(t)$ swept by its boundary. The shell is counted algebraically: where the boundary moves inward it removes volume. Add and subtract $\int_{V_f(t)} \phi(\vec{x}, t + dt)\, dV$ in the numerator of the limit definition (Eq. 3.2):

$$\int_{V_f(t+dt)} \phi(\vec{x}, t+dt)\, dV - \int_{V_f(t)} \phi(\vec{x}, t)\, dV = \int_{V_f(t)} \big[\phi(\vec{x}, t+dt) - \phi(\vec{x}, t)\big] dV + \int_{\delta V} \phi(\vec{x}, t+dt)\, dV$$

Eq. 3.2 rewritten as Eq. 3.3 (before the limit)

The first integral has a fixed domain $V_f(t)$ and measures how the field changes at fixed points. The second integral contains the field only over the thin shell and measures how much the domain grows.

Step 2: The unsteadiness term (Eq. 3.4)

Taylor-expand the integrand at a fixed point $\vec{x}$:

$$\phi(\vec{x}, t + dt) - \phi(\vec{x}, t) = \frac{\partial \phi}{\partial t}\, dt + O(dt^2)$$

Because ϕ is C^2 and $V_f(t)$ is bounded, the remainder is uniformly $O(dt^2)$ over the domain, so it can be integrated. Dividing by dt and letting $dt \to 0$ (the domain $V_f(t)$ does not depend on dt):

$$\lim_{dt \to 0} \frac{1}{dt} \int_{V_f(t)} \big[\phi(\vec{x}, t + dt) - \phi(\vec{x}, t)\big]\, dV = \int_{V_f(t)} \frac{\partial \phi}{\partial t}\, dV$$

Eq. 3.4

Step 3: The swept-volume term (Eq. 3.5)

A point $\vec{x}_s$ of the material surface moves in the time dt to $\vec{x}_s + \vec{v}\, dt + O(dt^2)$. Its displacement along the outward normal is $(\vec{v} \cdot \vec{n})\, dt$. The tangential part only slides the point along the surface and adds no volume. The shell above a surface element $d\sigma$ is therefore a prism of base $d\sigma$ and height $(\vec{v} \cdot \vec{n})\, dt$ (an oblique prism has the same volume as a right prism of equal base and height). Where $\vec{v} \cdot \vec{n} < 0$ the height is negative and the shell removes volume, which is the algebraic counting adopted in Step 1:

$$d(\delta V) = (\vec{v} \cdot \vec{n})\, dt\, d\sigma + O(dt^2)$$

Since $\phi(\vec{x}, t + dt) = \phi(\vec{x}, t) + O(dt)$, the integrand can be evaluated at time t at the cost of an $O(dt^2)$ error:

$$\int_{\delta V} \phi(\vec{x}, t + dt)\, dV = dt \int_{\Sigma_f(t)} \phi(\vec{x}, t)\, \vec{v} \cdot \vec{n}\, d\sigma + O(dt^2)$$

Dividing by dt and taking the limit gives the convective flux through the moving boundary:

$$\lim_{dt \to 0} \frac{1}{dt} \int_{\delta V} \phi(\vec{x}, t + dt)\, dV = \int_{\Sigma_f(t)} \phi\, \vec{v} \cdot \vec{n}\, d\sigma$$

Eq. 3.5

Step 4: Add both contributions (Eq. 3.6)

The limit of the sum is the sum of the limits, so Eqs. 3.4 and 3.5 give Eq. 3.6:

$$\frac{d}{dt}\int_{V_f(t)} \phi \, dV = \int_{V_f(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_f(t)} \phi \, \vec{v} \cdot \vec{n} \, d\sigma$$

Eq. 3.6

Step 5: Arbitrary moving control volume (Eq. 3.7)

A control volume $V_c(t)$ is a geometric region whose boundary points move with a prescribed velocity $\vec{v}_c(\vec{x}, t)$, not necessarily equal to the fluid velocity. Steps 1 to 4 used the velocity of the boundary only to compute the displacement $(\vec{v} \cdot \vec{n})\, dt$ of the surface. Repeating them with the boundary velocity $\vec{v}_c$ in place of $\vec{v}$ gives

$$\frac{d}{dt}\int_{V_c(t)} \phi \, dV = \int_{V_c(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_c(t)} \phi \, \vec{v}_c \cdot \vec{n} \, d\sigma$$

Eq. 3.7

Step 6: Subtraction at the instant of coincidence (Eq. 3.8)

Choose $V_c(t)$ so that at the instant t it occupies exactly the same region as $V_f(t)$, hence $\Sigma_c(t) = \Sigma_f(t)$ and the normal $\vec{n}$ is the same. (At later times the two volumes differ, which is why only the instantaneous values of the derivatives are compared.) The integrals of $\partial\phi/\partial t$ in Eqs. 3.6 and 3.7 are then identical. Subtracting Eq. 3.7 from Eq. 3.6:

$$\frac{d}{dt}\int_{V_f} \phi \, dV - \frac{d}{dt}\int_{V_c} \phi \, dV = \int_{\Sigma_c} \phi \, \vec{v} \cdot \vec{n} \, d\sigma - \int_{\Sigma_c} \phi \, \vec{v}_c \cdot \vec{n} \, d\sigma$$

Combining the two surface integrals gives Eq. 3.8:

$$\frac{d}{dt}\int_{V_f(t)} \phi \, dV = \frac{d}{dt}\int_{V_c(t)} \phi \, dV + \int_{\Sigma_c(t)} \phi \, (\vec{v} - \vec{v}_c) \cdot \vec{n} \, d\sigma$$

Eq. 3.8

Step 7: Checks (extension, not in the Notes)

One-dimensional check with the Leibniz rule. Take a one-dimensional "volume" $a(t) \le x \le b(t)$. The outward normals are $n = +1$ at $x = b$ and $n = -1$ at $x = a$, and the boundary velocities are $\dot{b}$ and $\dot{a}$. Eq. 3.7 gives $\int_a^b \partial\phi/\partial t \, dx + \phi(b,t)\dot{b} - \phi(a,t)\dot{a}$. Differentiating $F(t) = \int_{a(t)}^{b(t)} \phi(x,t)\, dx$ directly with the chain rule for $F(t,a,b)$ and Barrow's rule $\partial F/\partial b = \phi(b,t)$, $\partial F/\partial a = -\phi(a,t)$:

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial b}\dot{b} + \frac{\partial F}{\partial a}\dot{a} = \int_a^b \frac{\partial \phi}{\partial t}\, dx + \phi(b,t)\,\dot{b} - \phi(a,t)\,\dot{a}$$

Both results coincide, which confirms the sign convention of the flux term.

Dimensional check. If $[\phi]$ denotes the units of the density, then $[d\Phi/dt] = [\phi]\,\mathrm{m}^3/\mathrm{s}$, the unsteady term has units $[\phi]\,\mathrm{m}^3/\mathrm{s}$, and the flux term has units $[\phi] \cdot (\mathrm{m/s}) \cdot \mathrm{m}^2 = [\phi]\,\mathrm{m}^3/\mathrm{s}$.

Limiting case. If $\vec{v}_c = \vec{v}$ at the boundary, the relative velocity vanishes and Eq. 3.8 reduces to the identity $d\Phi_f/dt = d\Phi_c/dt$, as expected for a volume that moves with the fluid.

2. The Control Volume Toolbox: Selection Strategy

The entire art of integral fluid mechanics lies in choosing the optimal control volume V_c. The choice of V_c does not change the physical outcome, but a clever choice reduces pages of vector calculus to a 2-line algebraic equation.

Fixed Control Volume

$V_0, \vec{v}_c = \vec{0}$

Boundaries are fixed in the laboratory frame. Convective velocity is simply $\vec{v}$. Ideal for wind tunnels, stationary pipe bends, jet nozzles on testbenches, and steady heat exchangers.

Moving Control Volume

$V_c(t), \vec{v}_c \neq \vec{0}$

Translates at constant speed $\vec{v}_c$ or accelerates with a vehicle (rockets, cruising aircraft, traveling shock waves, hydraulic bores). Fluxes are carried by relative velocity $\vec{w} = \vec{v} - \vec{v}_c$.

Deformable Control Volume

$V_c(t), \vec{v}_c(\vec{x}, t)$

Boundary velocity varies along the control surface. Essential for reciprocating internal combustion engine pistons, inflating airbags, expanding rocket propellant bladders, and flexible vessels.

Boundary Decomposition and Sign Conventions

A control surface Σ_c is always partitioned into functional segments:

$$\Sigma_c = \Sigma_{\text{in}} \cup \Sigma_{\text{out}} \cup \Sigma_{\text{wall}} \cup \Sigma_{\text{free}}$$

- **Outward unit normal $\vec{n}$:** By universal convention, $\vec{n}$ is *strictly outward-pointing* from V_c at every boundary element $d\sigma$.
- **Inlets (Σ_{in}):** Fluid enters the volume against the normal vector: $(\vec{v} - \vec{v}_c) \cdot \vec{n} < 0$. This negative sign automatically ensures that inflow increases internal storage!
- **Outlets (Σ_{out}):** Fluid leaves the volume along the normal vector: $(\vec{v} - \vec{v}_c) \cdot \vec{n} > 0$.
- **Impermeable Walls (Σ_{wall}):** Fluid cannot cross a solid wall. Kinematically, $(\vec{v} - \vec{v}_c) \cdot \vec{n} = 0$, so convective flux through solid walls is identically zero.

3. The Moving Reference Frame Trick

One of the most powerful analytical techniques in aerospace engineering is **converting an unsteady flow problem into a steady one** by attaching the reference frame and control volume to a traveling disturbance.

Consider a solitary wave, a hydraulic bore, or a normal shock wave propagating at velocity $\vec{c} = c\hat{i}$. In the fixed laboratory frame, the fluid height, pressure, and velocity at any point x change rapidly as the front sweeps past ($\partial/\partial t \neq 0$), requiring complex partial differential equations.

The Trick: Attach a control volume V_c directly to the wave front, moving with constant velocity $\vec{v}_c = c\hat{i}$. Observed from this moving frame:

1. The wave front is completely stationary inside V_c.
2. The flow is entirely **steady**: $\frac{d}{dt}\int_{V_c} \rho b \, dV \equiv 0$.
3. Fluid enters from the right with relative speed $w_1 = v_1 - c$ and leaves to the left with relative speed $w_2 = v_2 - c$.

All time derivatives vanish, transforming the problem into a simple 1D steady flux balance!

CANONICAL PROBLEM WALKTHROUGH · CL2 SETUP

The Hydraulic Bore / Moving Shock Wave

Problem Statement: A tidal surge creates a hydraulic bore of height h_2 moving with constant wave velocity c into a calm, shallow channel of depth h_1 ($h_2 > h_1$) where water is initially at rest ($v_1 = 0$). Behind the bore, fluid moves forward with induced speed v_b. Determine the bore propagation speed c and induced velocity v_b in terms of h_1, h_2, and g.

Stage 1 **Select Moving Control Volume V_c**

Attach a 2D control volume of width W to the moving wave front, translating at velocity $\vec{v}_c = c\hat{i}$. In this frame, the flow is steady ($\partial/\partial t = 0$).

Calculate relative horizontal velocities:

- **Inflow (upstream ahead of bore):** Undisturbed water at rest ($v_1 = 0$) approaches the front with relative speed:

$$w_1 = v_1 - c = -c$$

- **Outflow (downstream behind bore):** Water moving at $v_b\hat{i}$ leaves the front with relative speed:

$$w_2 = v_b - c = -(c - v_b)$$

Stage 2 Apply Mass Conservation (Continuity)

Using the steady 1D continuity equation ($\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$):

$$\rho W h_1 |w_1| = \rho W h_2 |w_2| \implies h_1 c = h_2 (c - v_b)$$

Solving for the relative exit speed and induced speed v_b:

$$c - v_b = c\frac{h_1}{h_2} \implies v_b = c\left(1 - \frac{h_1}{h_2}\right) = c\frac{h_2 - h_1}{h_2}$$

Stage 3 Apply Linear Momentum Conservation

For a steady moving control volume with negligible bed friction over the short bore length:

$$\sum F_{x,\text{ext}} = \dot{m}(w_{2,x} - w_{1,x})$$

External horizontal forces are purely hydrostatic pressure distributions:

- Right face (height h_1): $F_{p1} = \int_0^{h_1} \rho g (h_1 - y) W \, dy = \frac{1}{2}\rho g W h_1^2$ (pushes left, $-\hat{i}$).
- Left face (height h_2): $F_{p2} = \int_0^{h_2} \rho g (h_2 - y) W \, dy = \frac{1}{2}\rho g W h_2^2$ (pushes right, $+\hat{i}$).

Net horizontal force (taking right as $+x$):

$$\sum F_x = \frac{1}{2}\rho g W h_2^2 - \frac{1}{2}\rho g W h_1^2 = \frac{1}{2}\rho g W (h_2^2 - h_1^2)$$

Net convective momentum flux leaving the volume (note $w_{2,x} = -(c - v_b)$ and $w_{1,x} = -c$):

$$\dot{m}(w_{2,x} - w_{1,x}) = (\rho W h_1 c)[-(c - v_b) - (-c)] = \rho W h_1 c v_b$$

Stage 4 Solve for Wave Speed c and Induced Velocity v_b

Equating net force to momentum flux:

$$\frac{1}{2}\rho g W (h_2^2 - h_1^2) = \rho W h_1 c v_b$$

Substitute $v_b = c\frac{h_2 - h_1}{h_2}$ into the right-hand side:

$$\frac{1}{2} g (h_2 - h_1)(h_2 + h_1) = h_1 c^2 \frac{h_2 - h_1}{h_2}$$

Dividing by $(h_2 - h_1) \neq 0$ yields the celebrated **Hydraulic Bore Propagation Speed Formula**:

$$\mathbf{c = \sqrt{g h_2 \frac{h_1 + h_2}{2 h_1}}}$$

And substituting back into v_b:

$$\mathbf{v_b = (h_2 - h_1)\sqrt{\frac{g(h_1 + h_2)}{2h_1h_2}}}$$

AEROSPACE EXAM INSIGHT: ACOUSTIC LIMIT

In the limit of an infinitesimal wave ($h_2 \to h_1$), $c \to \sqrt{gh_1}$, which is the shallow water gravity wave propagation speed (exact hydraulic analog of the acoustic speed of sound $a = \sqrt{\gamma RT}$). For finite jumps ($h_2 > h_1$), the bore is supersonic ($Fr = c/\sqrt{gh_1} > 1$), exactly analogous to a normal shock wave in compressible gas dynamics!